

Practical No 04

Aim:

Updating Documents in MongoDB Collections.

To update specific fields in existing MongoDB documents using update operators such as \$set, and to understand the use of updateOne() and updateMany() for modifying single and multiple documents.

Theory:

Updating stored data is an essential operation in MongoDB, allowing users to modify existing documents without replacing the entire document. MongoDB provides flexible update mechanisms that work efficiently with its document-oriented structure. The most commonly used update operator is \$set, which modifies the value of an existing field or creates the field if it does not already exist. This operator enables partial updates, meaning only the specified fields are altered while the rest of the document remains unchanged.

MongoDB offers two primary update functions:

1. **updateOne()** – Updates only the *first* document that matches the specified filter. This method is ideal when targeting a single, unique record based on a specific condition, such as updating a single student's age.
2. **updateMany()** – Updates *all* documents that match the filter condition. It is used for bulk updates, such as assigning a new attribute (e.g., scholarship eligibility) to multiple students in the same department.

When an update operation is performed, MongoDB identifies documents using the filter criteria and applies the update operators atomically. The database returns an acknowledgment indicating how many documents were matched and how many were modified. This ensures reliability even in concurrent environments. If a specified field does not previously exist in the document, \$set automatically adds it, demonstrating MongoDB's schema flexibility.

Commands:

```
db.students.updateOne( { name: "Pranjali" }, { $set: { age: 25 } } )  
db.students.updateMany( { dept: "ECE" }, { $set: { scholarship: false } } )
```

Expected Output:

Updated documents with modified fields.

Viva Questions:

1. What is an update operation in MongoDB?

An update operation modifies existing documents in a MongoDB collection without deleting or replacing the entire document.

2. What is the purpose of the \$set operator?

The \$set operator updates the value of a specified field or creates the field if it does not exist.

3. Difference between updateOne() and updateMany()?

- updateOne() updates only the first matching document.
- updateMany() updates all documents that match the given condition.

Conclusion:

In this practical, we successfully learned how to update existing documents in a MongoDB collection using update operators. By using the \$set operator along with updateOne() and updateMany() methods, we were able to modify specific fields without affecting the entire document.